1.2.12

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Question

For $|x| < 1$, $\coth(x)$ can be approximated as: (EC 2007)

a) x

b) x^2

c) $\frac{1}{x}$

d) $\frac{1}{x^2}$

Solution

Solution:
The hyperbolic cotangent function is defined as:

$$\coth(x) = \frac{\cosh(x)}{\sinh(x)} \tag{1}$$

To find the approximation for small x, we use the Taylor Series expansions for $\cosh(x)$ and $\sinh(x)$ centered at 0: Expansion of cosh(x) and sinh(x)

$$\cosh(x) : 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \tag{2}$$

$$\sinh(x) : x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \tag{3}$$

Now, substitute these into the $\coth(x)$ formula:

$$\coth(x) = \frac{1 + \frac{x^2}{2} + \dots}{x + \frac{x^3}{6} + \dots} \tag{4}$$

For very small values of x, we can focus on the dominant terms (the lowest power of x in the numerator and denominator):

$$\coth(x) \approx \frac{1}{x} \tag{5}$$

Correct Option: c) $\frac{1}{x}$

C Code

```c
#include <stdio.h>
#include <math.h>

double calculate_coth(double x) {
    if (x == 0.0) {
        printf("Error: coth(0) is undefined.\n");
        return NAN; // Returns 'Not a Number'
    }
    // Standard formula: cosh(x) / sinh(x)
    return cosh(x) / sinh(x);
}
```

C Code

```c
int main() {
    double x;

    printf("Enter the value of x: ");
    if (scanf("%lf", &x) != 1) {
        printf("Invalid input.\n");
        return 1;
    }
    double result = calculate_coth(x);

    if (!isnan(result)) {
        printf("coth(%.4f) = %.6f\n", x, result);
    }

    return 0;
}
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-2, 2, 400)
x = x[x != 0]

# Define the functions
coth_x = 1 / np.tanh(x)
approx_x = 1 / x

plt.figure(figsize=(10, 6))
plt.plot(x, coth_x, label=r'$\coth(x)$', color='blue', linewidth
    =2)
plt.plot(x, approx_x, '--', label=r'Approximation $\frac{1}{x}$',
     color='red', linewidth=2)
```

Python Code

```
plt.ylim(-10, 10)
plt.axhline(0, color='black',linewidth=0.5)
plt.axvline(0, color='black',linewidth=0.5)
plt.title(r'Comparison of $\coth(x)$ and its Approximation $\frac
    {1}{x}$', fontsize=14)
plt.xlabel('x', fontsize=12)
plt.ylabel('f(x)', fontsize=12)
plt.legend()
plt.grid(True, linestyle='--', alpha=0.7)

# Highlight the region |x| < 1
plt.fill_between(x, -10, 10, where=(np.abs(x) < 1), color='green'
    , alpha=0.1, label='Region $|x| < 1$')

plt.show()
```

Plot

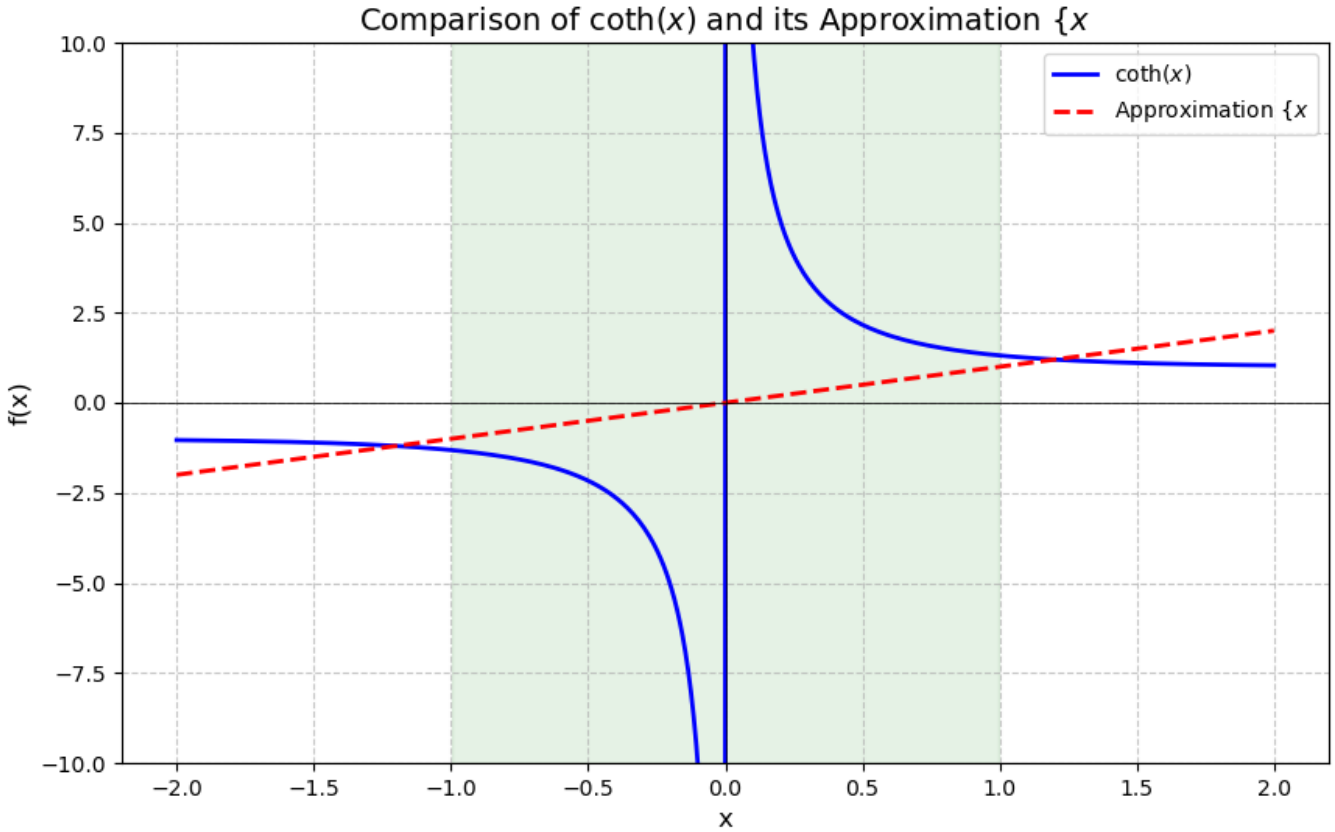

Figure: Coth(x) and x comparision for $|x| < 1$

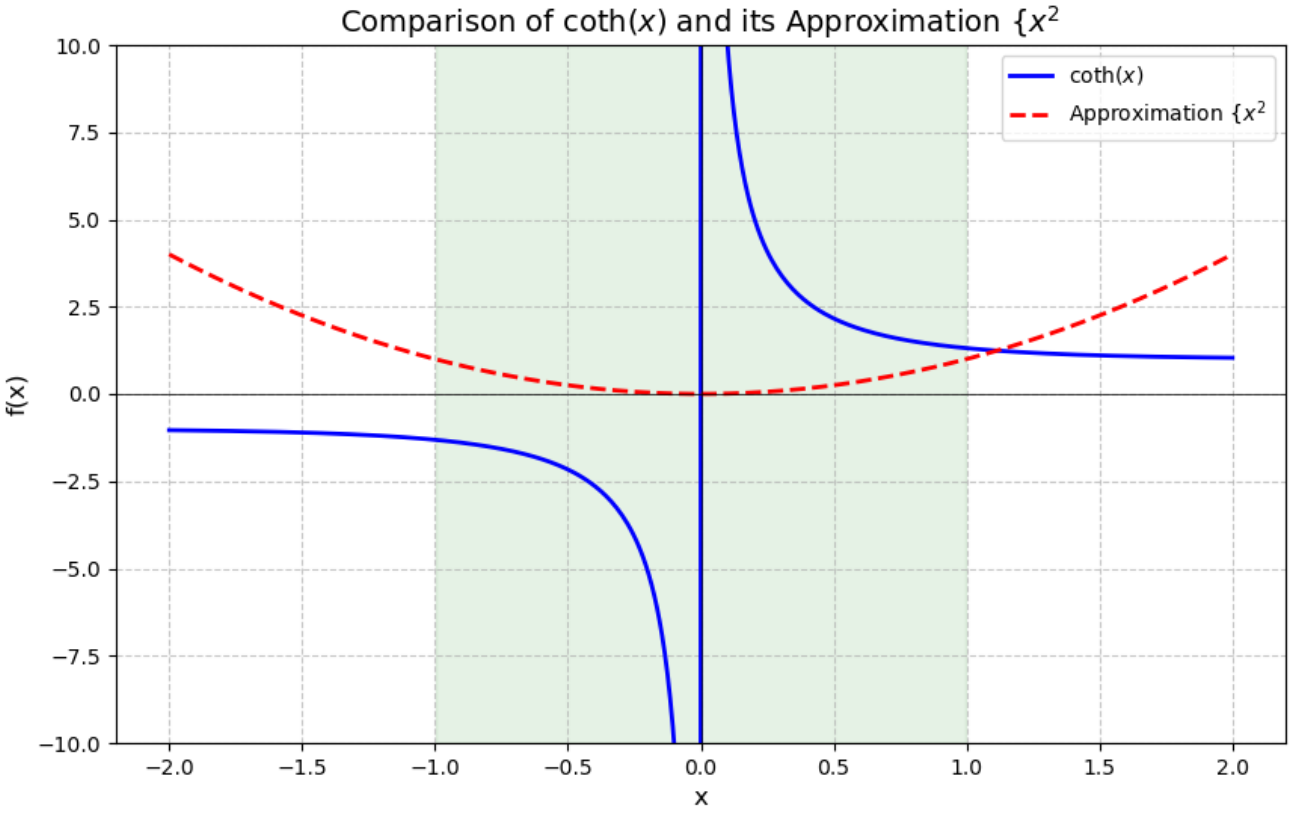

Figure: Coth(x) and x^2 comparision for $|x| < 1$

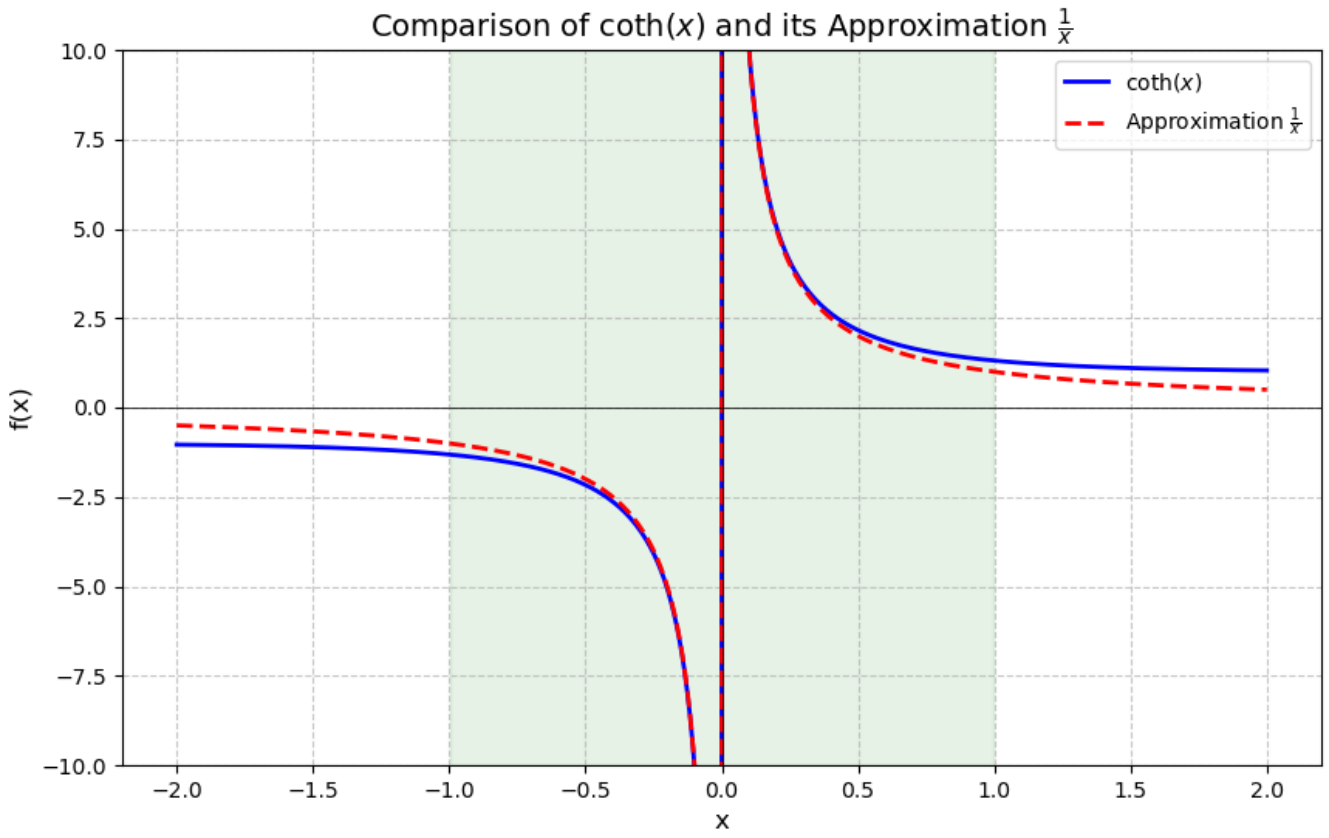

Figure: Coth(x) and 1/x comparision for $|x| < 1$

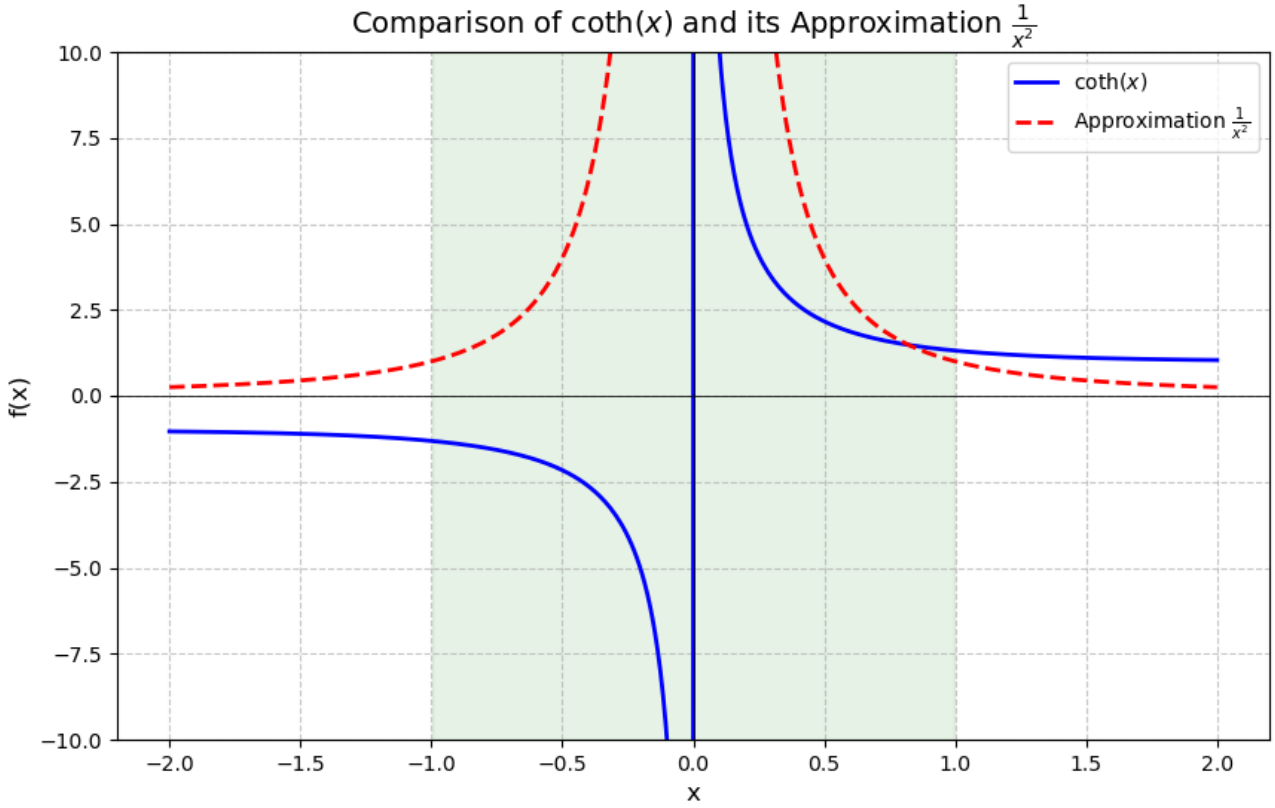

Figure: Coth(x) and $1/x^2$ comparision for $|x| < 1$